

Relatório 19 – Pipeline de Dados II

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1. Introdução

O objetivo desta atividade foi aprender sobre alguns conceitos de distribuição e organização de tarefas do Airflow com o curso “*Apache Airflow: The Hands-On Guide*” e desenvolver uma atividade prática. A tarefa consistiu em criar uma DAG que ingere diariamente metadados de *Papers* publicados no arXiv, em três categorias de pesquisa, processando-os em paralelo.

2. Desenvolvimento

Eu escolhi uma fonte real em vez de dados sintéticos, a API não cobra autenticação, o que diminuiu o escopo da atividade para focar nas orquestrações e não em gestão de credenciais, também, escolhi três categorias de Ciência da Computação com uma quantidade de arquivos diária suficiente para gerar dados reais e pequeno o bastante para não sobrecarregar a API durante os testes.

Foi feito task independente para cada categoria, elas são geradas de forma dinâmica a partir de uma lista do Python e eu escolhi isso porque uma task sequencial não usa o paralelismo do Airflow, o ganho de performance do LocalExecutor só aparece quando existe várias tasks independentes que o scheduler pode mandar ao mesmo tempo. Também obtive por usar o LocalExecutor, em vez do SequentialExecutor, só pra ver o paralelismo na prática.

Cada task de ingestão grava o resultado em um JSON e retorna só o caminho do arquivo, não a lista de *papers*, foi decidido isso porque na aula diz que o uso correto do XCom é para apenas guardar metadados de execução e não os dados em si. A task validar recebe os três caminhos via XCom e decide se o pipeline segue ou falha, isso sem nunca ter os dados brutos soltos fora dos arquivos gravados.

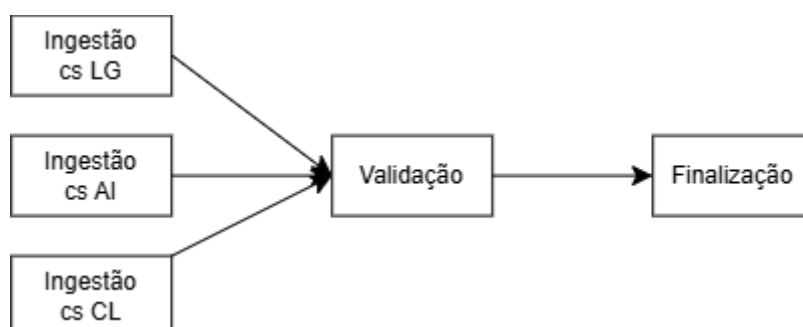


Figura 1. Diagrama do DAG. (autoria própria)



Figura 2. Fluxo de dados entre as tasks. (autoria própria)

Sobre os dados que são coletados, eu decidi as seguintes coisas:

Campo	Porque eu deixei ele
Titulo e resumo	São necessários para avaliar do que o <i>paper</i> se trata
Autores	Precisa para referenciar em pesquisas
Published e updated	Para avaliar o quão recente é o conteúdo
Primary_category e categories	Para filtrar ou analisar por área depois
Pdf_url	Ele é tirado do link com o <i>title="pdf"</i> no xml de resposta da API, é para usar para baixar o pdf em um etapa futura (um outro projeto meu usa eles, mas para o card é irrelevante e então deixei fora do escopo)
Categoria_busca	Necessário para procurar a origem de um <i>paper cross</i> listado em mais de uma categoria, antes de desduplicar

A desduplicação é usada porque um *paper* poder estar listado em outras categorias, sem tratar isso, a etapa de finalização teria o mesmo *paper* duas ou mais vezes, a etapa finalizar remove essa duplicidade usando o id do arXiv como chave. Já sobre o formato JSON e não CSV, basicamente é porque campos como autores e categories são listas, e no CSV, essas listas iriam ter que “achatar” em texto, o que vai fazer perder a estrutura, já no JSON permanece como lista nativas, isso vai ajudar em outras análises depois.

As prints que é pedida no card, da execução no docker CLI e o airflow:

```
Windows PowerShell
Copyright (C) Microsoft Corporation. Todos os direitos reservados.

Experimente a nova plataforma cruzada PowerShell https://aka.ms/pscore6

PS C:\Users\Edryck> cd "D:\Documents\BootCamp-LAMIA\III - Aprendizado de Máquina Aplicado\Card 19 - Pipelines de Dados II\Prática\card-arxiv"
PS D:\Documents\BootCamp-LAMIA\III - Aprendizado de Máquina Aplicado\Card 19 - Pipelines de Dados II\Prática\card-arxiv> ls

Diretório: D:\Documents\BootCamp-LAMIA\III - Aprendizado de Máquina Aplicado\Card 19 - Pipelines de Dados II\Prática\card-arxiv

Mode                LastWriteTime         Length Name
----                -
d-----          19/08/2026      19:38      data
d-----          19/08/2026      19:38      docker
d-----          19/08/2026      19:36      mnt
-a-----          19/08/2026      19:58           70 .env
-a-----          19/08/2026      19:58          179 .env.example
-a-----          19/08/2026      19:38           92 .gitignore
-a-----          19/08/2026      19:38        1539 docker-compose-LocalExecutor.yml

PS D:\Documents\BootCamp-LAMIA\III - Aprendizado de Máquina Aplicado\Card 19 - Pipelines de Dados II\Prática\card-arxiv> docker compose -f docker-compose-LocalExecutor.yml up airflow-init
(-) up 17/17
  Image postgres:15 Pulled
#1 [internal] load local bake definitions
#1 reading from stdin 767B 0.0s done
#1 DONE 0.0s
#2 [internal] load build definition from Dockerfile
#2 transferring dockerfile: 30B 0.0s
#2 transferring dockerfile: 169B 0.1s done
#2 DONE 0.2s
#3 [internal] load metadata for docker.io/apache/airflow:2.9.3-python3.11
#3 DONE 3.8s
#4 [internal] load .dockerignore
#4 transferring context: 2B 0.0s done
#4 DONE 0.1s
#5 [internal] load build context
#5 DONE 0.0s
#6 [1/3] FROM docker.io/apache/airflow:2.9.3-python3.11@sha256:cc5fcb91e93e4dfe4db1b53a9155dfa2670fb829891a9658a0f36ac55f67ef
#6 resolve docker.io/apache/airflow:2.9.3-python3.11@sha256:cc5fcb91e93e4dfe4db1b53a9155dfa2670fb829891a9658a0f36ac55f67ef 0.1s done
#6 ...

airflow-init-1 | [026-08-20T12:53:26.644+0000] [override.py:1858] INFO - Created Permission View: can read on My Password
airflow-init-1 | [026-08-20T12:53:26.659+0000] [override.py:1909] INFO - Added Permission can read on My Password to role Admin
airflow-init-1 | [026-08-20T12:53:26.659+0000] [override.py:1858] INFO - Created Permission View: can edit on My Profile
airflow-init-1 | [026-08-20T12:53:26.712+0000] [override.py:1909] INFO - Added Permission can edit on My Profile to role Admin
airflow-init-1 | [026-08-20T12:53:26.726+0000] [override.py:1858] INFO - Created Permission View: can read on My Profile
airflow-init-1 | [026-08-20T12:53:26.752+0000] [override.py:1909] INFO - Added Permission can read on My Profile to role Admin
airflow-init-1 | [026-08-20T12:53:26.821+0000] [override.py:1858] INFO - Created Permission View: can create on Users
airflow-init-1 | [026-08-20T12:53:26.840+0000] [override.py:1909] INFO - Added Permission can create on Users to role Admin
airflow-init-1 | [026-08-20T12:53:26.872+0000] [override.py:1858] INFO - Created Permission View: can delete on Users
airflow-init-1 | [026-08-20T12:53:26.898+0000] [override.py:1909] INFO - Added Permission can delete on Users to role Admin
airflow-init-1 | [026-08-20T12:53:26.917+0000] [override.py:1858] INFO - Created Permission View: can edit on Users
airflow-init-1 | [026-08-20T12:53:26.931+0000] [override.py:1909] INFO - Added Permission can edit on Users to role Admin
airflow-init-1 | [026-08-20T12:53:26.959+0000] [override.py:1858] INFO - Created Permission View: can delete on Users
airflow-init-1 | [026-08-20T12:53:26.975+0000] [override.py:1909] INFO - Added Permission can delete on Users to role Admin
airflow-init-1 | [026-08-20T12:53:27.015+0000] [override.py:1858] INFO - Created Permission View: menu access on List Users
airflow-init-1 | [026-08-20T12:53:27.030+0000] [override.py:1909] INFO - Added Permission menu access on List Users to role Admin
airflow-init-1 | [026-08-20T12:53:27.085+0000] [override.py:1858] INFO - Created Permission View: menu access on Security
airflow-init-1 | [026-08-20T12:53:27.087+0000] [override.py:1909] INFO - Added Permission menu access on Security to role Admin
airflow-init-1 | [026-08-20T12:53:27.136+0000] [override.py:1858] INFO - Created Permission View: can create on Roles
airflow-init-1 | [026-08-20T12:53:27.151+0000] [override.py:1909] INFO - Added Permission can create on Roles to role Admin
airflow-init-1 | [026-08-20T12:53:27.174+0000] [override.py:1858] INFO - Created Permission View: can read on Roles
airflow-init-1 | [026-08-20T12:53:27.189+0000] [override.py:1909] INFO - Added Permission can read on Roles to role Admin
airflow-init-1 | [026-08-20T12:53:27.211+0000] [override.py:1858] INFO - Created Permission View: can edit on Roles
airflow-init-1 | [026-08-20T12:53:27.226+0000] [override.py:1909] INFO - Added Permission can edit on Roles to role Admin
airflow-init-1 | [026-08-20T12:53:27.250+0000] [override.py:1858] INFO - Created Permission View: can delete on Roles
airflow-init-1 | [026-08-20T12:53:27.264+0000] [override.py:1909] INFO - Added Permission can delete on Roles to role Admin
airflow-init-1 | [026-08-20T12:53:27.308+0000] [override.py:1858] INFO - Created Permission View: menu access on List Roles
airflow-init-1 | [026-08-20T12:53:27.316+0000] [override.py:1909] INFO - Added Permission menu access on List Roles to role Admin
airflow-init-1 | [026-08-20T12:53:27.366+0000] [override.py:1858] INFO - Created Permission View: can read on User Stats Chart
airflow-init-1 | [026-08-20T12:53:27.381+0000] [override.py:1909] INFO - Added Permission can read on User Stats Chart to role Admin
airflow-init-1 | [026-08-20T12:53:27.423+0000] [override.py:1858] INFO - Created Permission View: menu access on User's Statistics
airflow-init-1 | [026-08-20T12:53:27.437+0000] [override.py:1909] INFO - Added Permission menu access on User's Statistics to role Admin
airflow-init-1 | [026-08-20T12:53:27.496+0000] [override.py:1858] INFO - Created Permission View: can read on Permissions
airflow-init-1 | [026-08-20T12:53:27.513+0000] [override.py:1909] INFO - Added Permission can read on Permissions to role Admin
airflow-init-1 | [026-08-20T12:53:27.562+0000] [override.py:1858] INFO - Created Permission View: menu access on Actions
airflow-init-1 | [026-08-20T12:53:27.581+0000] [override.py:1909] INFO - Added Permission menu access on Actions to role Admin
airflow-init-1 | [026-08-20T12:53:27.657+0000] [override.py:1858] INFO - Created Permission View: can read on View Menus
airflow-init-1 | [026-08-20T12:53:27.673+0000] [override.py:1909] INFO - Added Permission can read on View Menus to role Admin
airflow-init-1 | [026-08-20T12:53:27.716+0000] [override.py:1858] INFO - Created Permission View: menu access on Resources
airflow-init-1 | [026-08-20T12:53:27.735+0000] [override.py:1909] INFO - Added Permission menu access on Resources to role Admin
airflow-init-1 | [026-08-20T12:53:27.813+0000] [override.py:1858] INFO - Created Permission View: can read on Permission Views
airflow-init-1 | [026-08-20T12:53:27.828+0000] [override.py:1909] INFO - Added Permission can read on Permission Views to role Admin
airflow-init-1 | [026-08-20T12:53:27.876+0000] [override.py:1858] INFO - Created Permission View: menu access on Permission Pairs
airflow-init-1 | [026-08-20T12:53:28.054+0000] [override.py:1909] INFO - Added Permission menu access on Permission Pairs to role Admin
airflow-init-1 | [026-08-20T12:53:28.054+0000] [override.py:1543] INFO - Added user admin
airflow-init-1 | User "admin" created with role "Admin"
airflow-init-1 exited with code 0
PS D:\Documents\BootCamp-LAMIA\III - Aprendizado de Máquina Aplicado\Card 19 - Pipelines de Dados II\Prática\card-arxiv>
```

Figura 3 e 4. Criando a imagem e confirmação de sucesso. (autoria própria)

```

Windows PowerShell
PS D:\Documentos\BootCamp-LAMIA\III - Aprendizado de Máquina Aplicado\Card 19 - Pipelines de Dados II\Prática\card-arxiv> docker compose -f docker-compose-LocalExecutor.yml up -d
#1 [internal] load local bake definitions
#1 reading from stdin 2.08kB 0.0s done
#1 DONE 0.0s

#2 [scheduler internal] load build definition from Dockerfile
#2 transferring dockerfile: 169B 0.0s done
#2 DONE 0.1s

#3 [webserver internal] load metadata for docker.io/apache/airflow:2.9.3-python3.11
#3 DONE 1.4s

#4 [webserver internal] load .dockerignore
#4 transferring context: 2B 0.0s done
#4 DONE 0.0s

#5 [webserver 1/3] FROM docker.io/apache/airflow:2.9.3-python3.11@sha256:cc5fcb91e93e4dfe4fd8b1b53a9155dfe2670fb829891a9658a0f36ac55f67ef
#5 resolve docker.io/apache/airflow:2.9.3-python3.11@sha256:cc5fcb91e93e4dfe4fd8b1b53a9155dfe2670fb829891a9658a0f36ac55f67ef 0.1s done
#5 DONE 0.1s

#6 [webserver internal] load build context
#6 transferring context: 37B done
#6 DONE 0.0s

#7 [scheduler 2/3] COPY requirements.txt /requirements.txt
#7 CACHED

#8 [scheduler 3/3] RUN pip install --no-cache-dir -r /requirements.txt
#8 CACHED

#7 [webserver 2/3] COPY requirements.txt /requirements.txt
#7 CACHED

#9 [scheduler] exporting to image
#9 exporting layers 0.0s done
#9 exporting manifest sha256:678e5ae195bd478f6df966e19bc0a69ebdf78129c04b0d21a5d7b7adb1b63ef 0.0s done
#9 exporting config sha256:e70695a6cd284856249e43e7028d5dfe07d64bf82719de2c2dd4bed25baf9a2
#9 exporting config sha256:e70695a6cd284856249e43e7028d5dfe07d64bf82719de2c2dd4bed25baf9a2 0.1s done
#9 exporting attestation manifest sha256:e444b3aa53992a5e2be8d4c62f928b0ef0349402ee9d7b4b75615e21afabc716
#9 exporting attestation manifest sha256:e444b3aa53992a5e2be8d4c62f928b0ef0349402ee9d7b4b75615e21afabc716 0.1s done
#9 exporting manifest list sha256:eb6938473badd945be141f716846ec371f01e38fc7caef2c7ee83df143ec176
#9 exporting manifest list sha256:eb6938473badd945be141f716846ec371f01e38fc7caef2c7ee83df143ec176 0.1s done
#9 naming to docker.io/library/card-arxiv-scheduler:latest 0.0s done
#9 unpacking to docker.io/library/card-arxiv-scheduler:latest 0.1s done
#9 DONE 0.0s

#10 [webserver] exporting to image
#10 exporting layers 0.0s done
#10 exporting manifest sha256:80935f9670c82d7c3d134ac8e355777b0cd7ade5d51b0d7ff920a5b99ff0b304 0.1s done
#10 exporting manifest sha256:80935f9670c82d7c3d134ac8e355777b0cd7ade5d51b0d7ff920a5b99ff0b304 0.1s done

Windows PowerShell
#7 [scheduler 2/3] COPY requirements.txt /requirements.txt
#7 CACHED

#8 [scheduler 3/3] RUN pip install --no-cache-dir -r /requirements.txt
#8 CACHED

#7 [webserver 2/3] COPY requirements.txt /requirements.txt
#7 CACHED

#9 [scheduler] exporting to image
#9 exporting layers 0.0s done
#9 exporting manifest sha256:678e5ae195bd478f6df966e19bc0a69ebdf78129c04b0d21a5d7b7adb1b63ef 0.0s done
#9 exporting config sha256:e70695a6cd284856249e43e7028d5dfe07d64bf82719de2c2dd4bed25baf9a2
#9 exporting config sha256:e70695a6cd284856249e43e7028d5dfe07d64bf82719de2c2dd4bed25baf9a2 0.1s done
#9 exporting attestation manifest sha256:e444b3aa53992a5e2be8d4c62f928b0ef0349402ee9d7b4b75615e21afabc716
#9 exporting attestation manifest sha256:e444b3aa53992a5e2be8d4c62f928b0ef0349402ee9d7b4b75615e21afabc716 0.1s done
#9 exporting manifest list sha256:eb6938473badd945be141f716846ec371f01e38fc7caef2c7ee83df143ec176
#9 exporting manifest list sha256:eb6938473badd945be141f716846ec371f01e38fc7caef2c7ee83df143ec176 0.1s done
#9 naming to docker.io/library/card-arxiv-scheduler:latest 0.0s done
#9 unpacking to docker.io/library/card-arxiv-scheduler:latest 0.1s done
#9 DONE 0.0s

#10 [webserver] exporting to image
#10 exporting layers 0.0s done
#10 exporting manifest sha256:80935f9670c82d7c3d134ac8e355777b0cd7ade5d51b0d7ff920a5b99ff0b304 0.1s done
#10 exporting config sha256:d5be4a02b3d8ea50a5620e16ba4d92fbc1f58e69e5b27bc182abe0f3379bf4 0.1s done
#10 exporting attestation manifest sha256:160980209daa2924e0491d4e74a144c9c85375bdc00a9c4ad3f6555375ddaf8 0.1s done
#10 exporting manifest list sha256:d4bed9d34f51c27ef3225fdbc4159193e0ad077a9d925e58c98fe18f8c32592 0.1s done
#10 exporting manifest list sha256:d4bed9d34f51c27ef3225fdbc4159193e0ad077a9d925e58c98fe18f8c32592 0.1s done
#10 naming to docker.io/library/card-arxiv-webserver:latest 0.0s done
#10 unpacking to docker.io/library/card-arxiv-webserver:latest 0.1s done
#10 DONE 0.0s

#11 [scheduler] resolving provenance for metadata file
#11 DONE 0.2s

#12 [webserver] resolving provenance for metadata file
#12 DONE 0.1s

[+] up 0/0
Image card-arxiv-webserver Built 5.1s
Image card-arxiv-scheduler Built 5.1s
Container card-arxiv-postgres-1 Healthy 1.1s
Container card-arxiv-webserver-1 Started 2.1s
Container card-arxiv-scheduler-1 Started 2.1s
Container card-arxiv-airflow-init-1 Started 0.0s
PS D:\Documentos\BootCamp-LAMIA\III - Aprendizado de Máquina Aplicado\Card 19 - Pipelines de Dados II\Prática\card-arxiv>

PS D:\Documentos\BootCamp-LAMIA\III - Aprendizado de Máquina Aplicado\Card 19 - Pipelines de Dados II\Prática\card-arxiv> docker ps
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS NAMES
d60361ed1594 card-arxiv-webserver "/usr/bin/dumb-init ..." About a minute ago Up About a minute 0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp card-arxiv-webserver-1
b91836a34aa2 card-arxiv-scheduler "/usr/bin/dumb-init ..." About a minute ago Up About a minute 8080/tcp card-arxiv-scheduler-1
4887f7c14d0a postgres:15 "docker-entrypoint.s..." 3 minutes ago Up 3 minutes (healthy) 5432/tcp card-arxiv-postgres-1
1ddc0a9cf17e flown-backend "docker-entrypoint.s..." 22 hours ago Up 21 hours 0.0.0.0:3333->3333/tcp, [::]:3333->3333/tcp flown-backend-1
PS D:\Documentos\BootCamp-LAMIA\III - Aprendizado de Máquina Aplicado\Card 19 - Pipelines de Dados II\Prática\card-arxiv>

```

Figura 5, 6 e 7. Subindo o conateinar e conferindo se deu certo. (autoria própria)

```
Windows PowerShell
PS D:\Documentos\BootCamp-LAMIA\III - Aprendizado de Máquina Aplicado\Card 19 - Pipelines de Dados II\Prática\card-arxiv> docker compose -f docker-compose-LocalExecutor.yml exec webserver airf
low tasks test arxiv_dag ingest_cs_LG 2026-08-01
[2026-08-20T13:14:01.002+0000] {dagbag.py:545} INFO - Filling up the DagBag from /opt/***/dags
[2026-08-20T13:14:01.020+0000] {taskinstance.py:2076} INFO - Dependencies all met for dep_context=non-requeueable deps ti=<TaskInstance: arxiv_dag.ingest_cs_LG ____temporary_run_2026-08-20T1
3:14:01.395795+00:00____ [None]>
[2026-08-20T13:14:01.711+0000] {taskinstance.py:2076} INFO - Dependencies all met for dep_context=queueable deps ti=<TaskInstance: arxiv_dag.ingest_cs_LG ____temporary_run_2026-08-20T13:14
:01.395795+00:00____ [None]>
[2026-08-20T13:14:01.711+0000] {taskinstance.py:2386} INFO - Starting attempt 1 of 4
[2026-08-20T13:14:01.717+0000] {taskinstance.py:2388} WARNING - cannot record queued_duration for task ingest_cs_LG because previous state change time has not been saved
[2026-08-20T13:14:01.726+0000] {taskinstance.py:2330} INFO - Executing <Task(PythonOperator): ingest_cs_LG> on 2026-08-01 00:00:00+00:00
[2026-08-20T13:14:02.104+0000] {taskinstance.py:2648} INFO - Exporting env vars: AIRFLOW_CTX_DAG_OWNER='edryck' AIRFLOW_CTX_DAG_ID='arxiv_dag' AIRFLOW_CTX_TASK_ID='ingest_cs_LG' AIRFLOW_CTX_EX
ECUTION_DATE='2026-08-01T00:00:00+00:00' AIRFLOW_CTX_TRY_NUMBER='1' AIRFLOW_CTX_DAG_RUN_ID='____temporary_run_2026-08-20T13:14:01.395795+00:00____'
[2026-08-20T13:14:02.257+0000] {taskinstance.py:490} INFO - ::endgroup::
[2026-08-20T13:14:02.257+0000] {__init__.py:565} INFO - Requesting page (first: True, try: 0): https://export.arxiv.org/api/query?search_query=cat%3Acs.LG+AND+submittedDate%3A%5B202608010000+T
0+202608012359%5D&id_list=&sortBy=submittedDate&sortOrder=ascending&start=0&max_results=100
[2026-08-20T13:14:02.497+0000] {__init__.py:497} INFO - Got first page: 66 of 66 total results
[2026-08-20T13:14:02.791+0000] {__init__.py:237} INFO - Done. Returned value was: /opt/***/data/raw/2026-08-01/cs.LG.json
[2026-08-20T13:14:02.792+0000] {taskinstance.py:441} INFO - ::group::Post task execution logs
[2026-08-20T13:14:02.792+0000] {taskinstance.py:1206} INFO - Marking task as SUCCESS. dag_id=arxiv_dag, task_id=ingest_cs_LG, run_id=____temporary_run_2026-08-20T13:14:01.395795+00:00____, exe
cution_date=20260801T000000, start_date=, end_date=20260801T131402
PS D:\Documentos\BootCamp-LAMIA\III - Aprendizado de Máquina Aplicado\Card 19 - Pipelines de Dados II\Prática\card-arxiv>
```

Figura 8. Testando umas das tasks. (autoria própria)

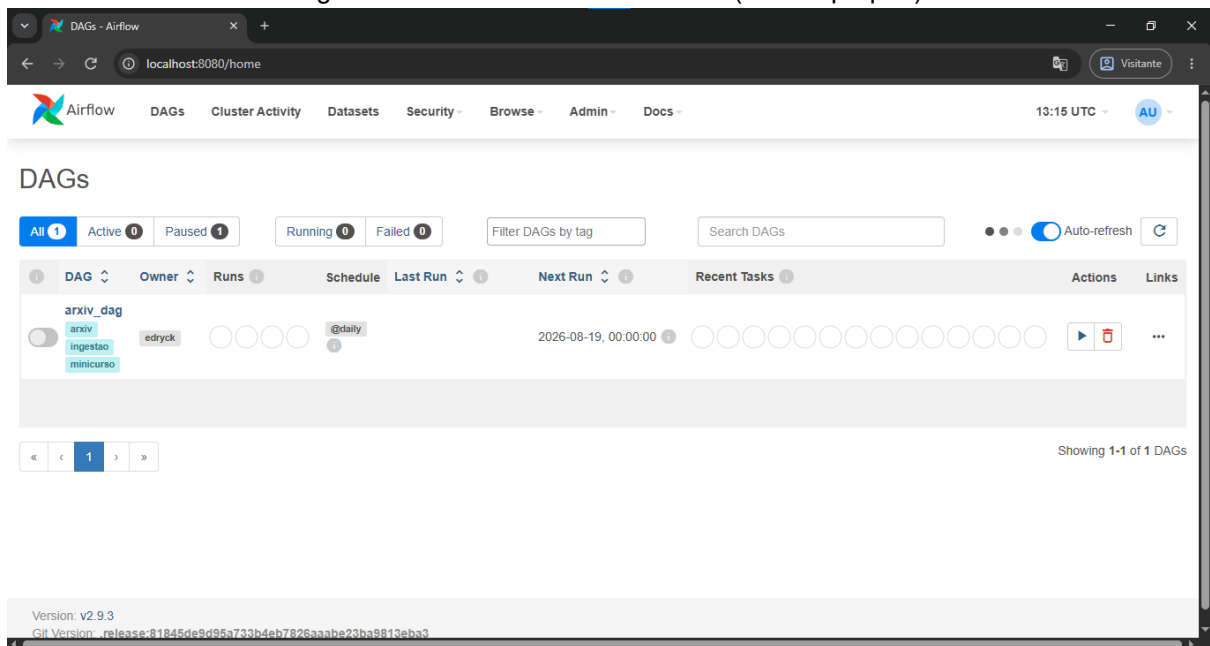


Figura 9. Colocando o DAG para rodar no Airflow. (autoria própria)

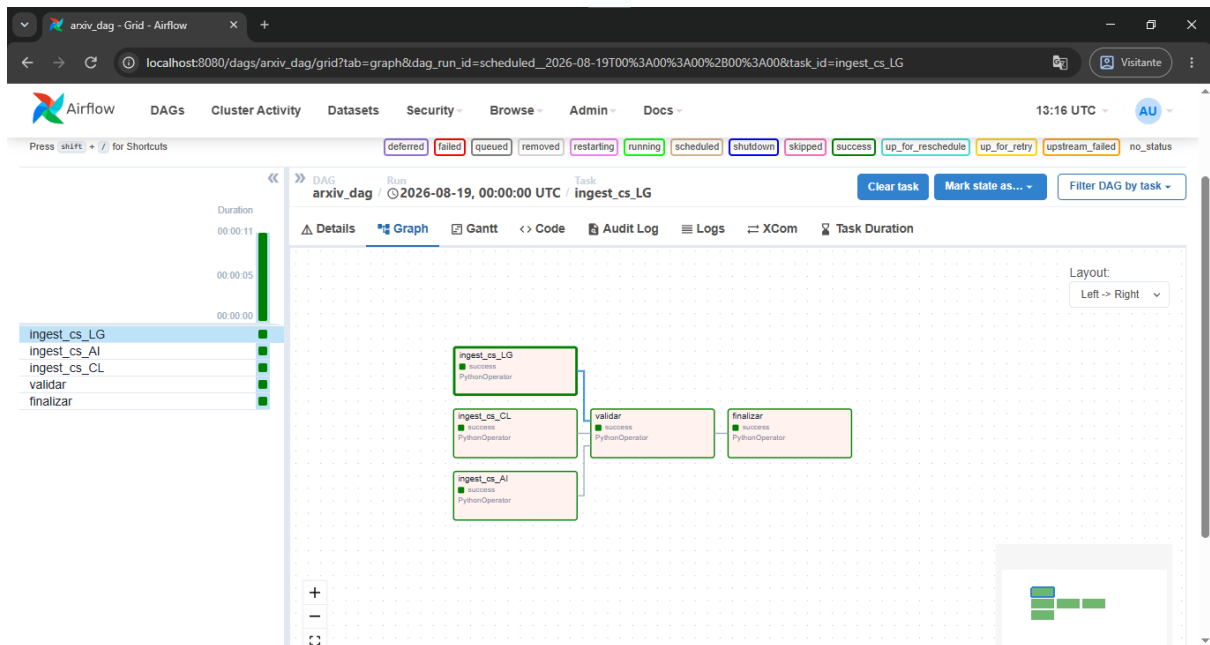


Figura 10. Visualização em grafo do DAG. (autoria própria)

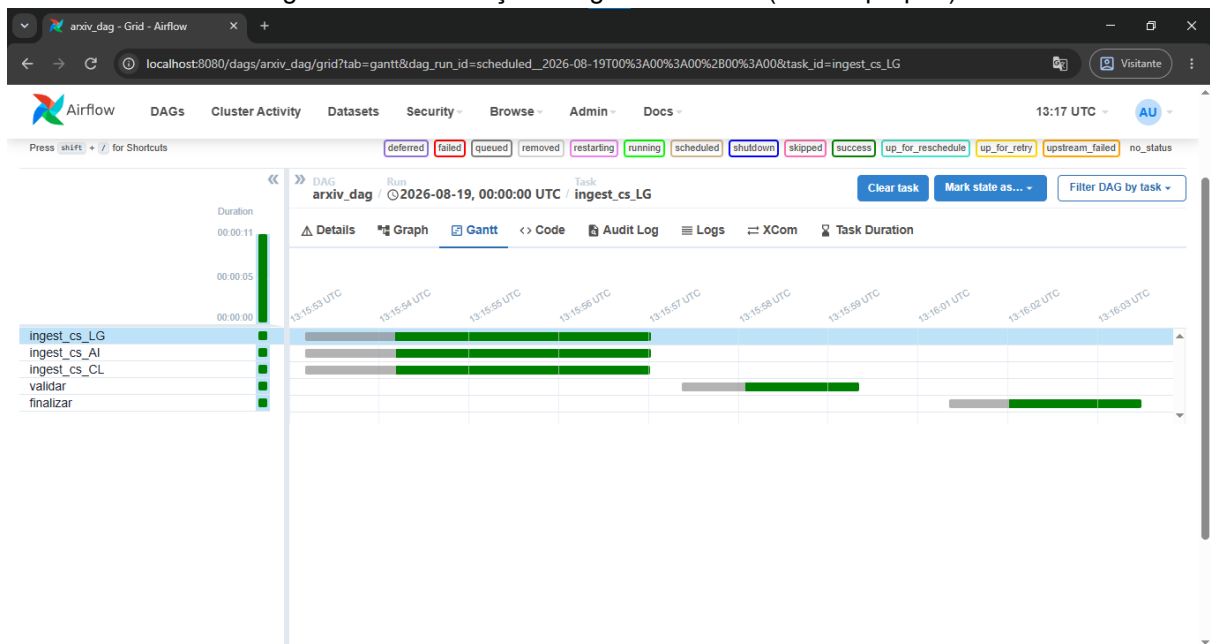


Figura 11. Visualização por tempo de execução das tasks. (autoria própria)

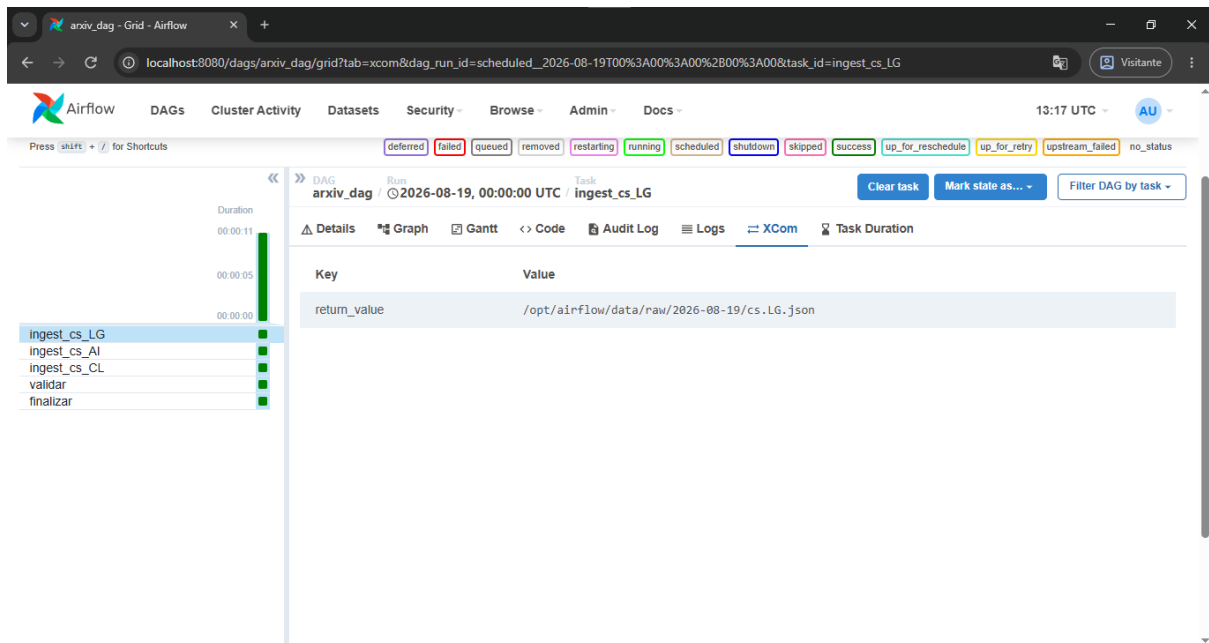


Figura 12. Metadados retornados ao final do DAG executado com sucesso. (autoria própria)

3. Conclusão

O Airflow separa a definição do fluxo, a DAG, da execução distribuída dele, ou seja, a mesma DAG que roda sequencialmente com o SequentialExecutor consegue executar as três ingestões em paralelo só trocando a configuração do executor, isso sem alterar o código de negócio. Também o XCom não deve ser tratado como um canal de transporte de dados e sim metadados, nesse pipeline significa passar os caminhos dos arquivos entre as tasks e não os dados em si. Mas a principal coisa que aprendi foi que a robustez de um pipeline de dados depende mais de decisões estruturais simples, como onde os dados são particionados, o que trafega entre as tasks e como duplicidade e falhas são tratadas, do que do quão complexo o código tem que ser.

4. Referências

Curso: *Apache Airflow: The Hands-On Guide*, material das seções 5 a 9.

Documentação da API do arXiv, disponível em: <https://info.arxiv.org/help/api/user-manual.html>.

Página inicial da API do arXiv, disponível em: <https://info.arxiv.org/help/api/>.